PIPELINE AUDIT AND HEALTH REPORT

Healthcare: emergency department

Patients arrive as minor, urgent or critical. A triage nurse (priority queue, walk-outs when the waiting room is full) sends critical patients to a trauma bay and everyone else to the least busy of four exam rooms. Urgent cases often need the lab before discharge. Watch triage and the exam rooms as arrivals rise.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 21 | 10 replications

HEALTH SCORE

83 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 1 | Low 2 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
12.4 units/h	95 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
13.5 units	3336 s	exam_1

Executive summary

The line keeps up with demand (0.00% lost, sources never throttled): throughput equals the arrival rate, so nothing is binding right now. The busiest station is 'exam_1' at 99% busy; that is where capacity runs out first as demand grows.

What to fix first

- High (exam_1): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'exam_1' (+1.5% throughput).
- Medium (steady state): Lengthen the run or use a warm-up period before trusting averages.
- Low (failure model): Add realistic MTBF and MTTR to the stations that break down in practice.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

Best measured improvement

Best lever: 10% faster processing at 'exam_1' -> +0.2 units/hour (1.5%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**exam_1**	The bottleneck is 99% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'exam_1' (+1.5% throughput).	-8
Medium	Model	**steady state**	Little's Law differs by 14.8%: the run has not reached steady state (units are still accumulating). Action: Lengthen the run or use a warm-up period before trusting averages.	-4
Low	Reliability	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic. Action: Add realistic MTBF and MTTR to the stations that break down in practice.	-3
Low	Model	**run length**	The measured window (7.5 h) is only 8x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**exam_1**	Highest utilisation in the line (99%); the line currently keeps up with demand. Action: This is where capacity runs out first as demand grows.	
Info	Reliability	**dependencies**	1 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
arrivals	source	-	-	-	-	-
triage	machine		73.1	0.3	0.0	1.4 / 5
trauma	machine		56.2	0.0	0.0	0.6 / 2
exam_1 bottleneck	machine		99.2	0.0	0.0	2.7 / 5
exam_2	machine		99.1	0.0	0.0	2.5 / 5
exam_3	machine		92.1	0.0	0.0	2.2 / 5
exam_4	machine		71.3	0.0	0.0	1.8 / 5
lab	machine		61.6	0.0	0.0	1.0 / 3
discharge	machine		74.5	0.0	0.0	1.3 / 5
home	sink	-	-	-	-	-
admitted	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	12.45 units/h	+- 0.62	0.86	5.0%
Average WIP	18.23 units	+- 4.78	6.68	26.2%
Cycle time	4191.16 s	+- 790.39	1104.97	18.9%
Arrivals lost	2.19 %	+- 2.56	3.58	116.6%

No replication deadlocked.

Failure model

No machine has a failure model.

Single points of failure

All units must pass through: triage. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 119 = completed 95 + scrapped 0 + lost 0 + in system 24
Little's Law	FAIL	L = 13.48, lambda x W = 11.49 (gap 14.8%)
Run length	WARN	window = 8 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 11.9 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
exam_1	10% faster processing	12.0	**+0.2**	+1.5%
exam_3	10% faster processing	12.0	**+0.2**	+1.5%
exam_2	10% faster processing	12.0	**+0.1**	+1.1%
exam_1	+2 buffer slots	11.9	**+0.0**	+0.0%
exam_2	+2 buffer slots	11.9	**+0.0**	+0.0%
exam_3	+2 buffer slots	11.9	**+0.0**	+0.0%

Best lever: 10% faster processing at 'exam_1' -> +0.2 units/hour (1.5%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
arrivals	source	interarrival=exponential(240), mix={"critical":0.1,"minor":0.55,"urgent":0.35}, priority={"critical":3,"minor":1,"urgent":2}
triage	machine	capacity=10, discipline=priority, on_full=drop, process=normal(180,40)
trauma	machine	capacity=3, discipline=fifo, process=normal(1500,300)
exam_1	machine	capacity=4, discipline=priority, process_by_type={"minor":{"dist":"normal","mean":900,"stddev":200},"urgent":{"dist":"normal","mean":1500,"stddev":300}}
exam_2	machine	capacity=4, discipline=priority, process_by_type={"minor":{"dist":"normal","mean":900,"stddev":200},"urgent":{"dist":"normal","mean":1500,"stddev":300}}
exam_3	machine	capacity=4, discipline=priority, process_by_type={"minor":{"dist":"normal","mean":900,"stddev":200},"urgent":{"dist":"normal","mean":1500,"stddev":300}}
exam_4	machine	capacity=4, discipline=priority, process_by_type={"minor":{"dist":"normal","mean":900,"stddev":200},"urgent":{"dist":"normal","mean":1500,"stddev":300}}
lab	machine	capacity=6, process=exponential(600)
discharge	machine	capacity=6, process=normal(240,60)
home	sink	(defaults)
admitted	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**exam_1**	The bottleneck is 99% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**steady state**	Little's Law differs by 14.8%: the run has not reached steady state (units are still accumulating).	-4
Low	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic.	-3
Low	**run length**	The measured window (7.5 h) is only 8x the cycle time.	-2
	Total deductions		**-17**
	Health score		**83**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.